



SIMATS Online Programming Lab

JAVA PROGRAMMING



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CO1-AT3-Coding Challenge

View Only

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QUESTIONS

Q1

Smart University Payroll
Management System
25 marks



Q2

Smart Vehicle Insurance Claim
System
25 marks



2/2 answered

Q2 Smart Vehicle Insurance Claim System 25 marks

Problem Statement An insurance company processes claims for three categories of vehicles: Car, Bike, and Truck. The claim amount depends on the vehicle type. **Requirements** Superclass: Vehicle Private members: vehicleNumber ownerName insuredAmount Subclasses: Car Bike Truck Each subclass overrides calculateClaimAmount() using different formulas. The application should: Read details for N vehicles Store them using an array of superclass references Calculate claim amounts using runtime polymorphism Search by vehicle number Display the vehicle with the highest claim amount

Question 3 (Very Hard) Online Shopping Order Processing System (50 Marks) **Problem Statement** An e-commerce company sells products in three categories: Electronics Clothing Grocery The billing process varies for each category. **Requirements** Superclass: Product Private members: productId productName price quantity Subclasses: Electronics Clothing Grocery Each subclass overrides calculateBill(). The program should: Store products in an array of superclass references Calculate the final bill using runtime polymorphism Search products by ID Display the most expensive order Compute total revenue

Claim Formula (Recommended)

Car

$$\text{Claim} = \text{insuredAmount} \times 0.80$$

Bike

$$\text{Claim} = \text{insuredAmount} \times 0.60$$

Truck

$$\text{Claim} = \text{insuredAmount} \times 0.90$$

Test Case 1

Input

3

Car

TN10AB1234

Ramesh

800000

Bike

TN09XY5678

Suresh

100000

Truck

TN11TR9999

Logan

1500000

Search Vehicle:

TN09XY5678

Expected Output

Vehicle : TN10AB1234

Claim Amount : 640000.0

Vehicle : TN09XY5678

Claim Amount : 60000.0

Vehicle : TN11TR9999

Claim Amount : 1350000.0

Highest Claim Vehicle

TN11TR9999

1350000.0

Vehicle Found

Owner : Suresh

Claim Amount : 60000.0

Your Code

```
1 import java.util.*;
2
3 class Main
4 {
5     public static String readInput(Scanner
6     {
7         String s="";
8
9         while(sc.hasNextLine())
10        {
11            s=sc.nextLine().trim();
12
13            if(!s.isEmpty())
14                return s;
15        }
16
17        return s;
18    }
19
20    public static void main(String args[])
21    {
22        Scanner sc=new Scanner(System.in)
23
24        int n=Integer.parseInt(readInput(
25
26        Vehicle vehicle[]=new Vehicle[n];
27
28        for(int i=0;i<n;i++)
29        {
30            String type=readInput(sc);
31
32            String vehicleNumber=readInpu
33            String ownerName=readInput(sc
34            double insuredAmount=Double.p
35
36            if(type.equalsIgnoreCase("Car
37            {
38                vehicle[i]=new Car(vehicl
39            }
40            else if(type.equalsIgnoreCase
41            {
```

```
42         vehicle[i]=new Bike(vehic
43     }
44     else if(type.equalsIgnoreCase
45     {
46         vehicle[i]=new Truck(vehi
47     }
48 }
49
50 String search=readInput(sc);
51
52 if(search.equalsIgnoreCase("Searc
53 {
54     search=readInput(sc);
55 }
56
57 Vehicle highest=vehicle[0];
58
59 for(int i=0;i<n;i++)
60 {
61     if(vehicle[i].calculateClaimA
62     {
63         highest=vehicle[i];
64     }
65 }
66
67 for(int i=0;i<n;i++)
68 {
69     System.out.println("Vehicle :
70     System.out.println("Claim Amc
71     System.out.println();
72 }
73
74 System.out.println("Highest Claim
75 System.out.println(highest.getVeh
76 System.out.println(highest.calcul
77 System.out.println();
78
79 boolean found=false;
80
81 for(int i=0;i<n;i++)
82 {
83     if(vehicle[i].getVehicleNumbe
84     {
85         System.out.println("Vehic
86         System.out.println("Owner
87         System.out.println("Claim
88
89         found=true;
```

```
90         break;
91     }
92 }
93
94     if(!found)
95     {
96         System.out.println("Vehicle N
97     }
98 }
99 }
100
101 class Vehicle
102 {
103     private String vehicleNumber;
104     private String ownerName;
105     private double insuredAmount;
106
107     public Vehicle()
108     {
109
110     }
111
112     public Vehicle(String vehicleNumber,S
113     {
114         this.vehicleNumber=vehicleNumber;
115         this.ownerName=ownerName;
116         this.insuredAmount=insuredAmount;
117     }
118
119     public String getVehicleNumber()
120     {
121         return vehicleNumber;
122     }
123
124     public String getOwnerName()
125     {
126         return ownerName;
127     }
128
129     public double getInsuredAmount()
130     {
131         return insuredAmount;
132     }
133
134     public double calculateClaimAmount()
135     {
136         return insuredAmount;
137     }
```

```
138 }
139 class Car extends Vehicle
140 {
141     public Car()
142     {
143
144     }
145
146     public Car(String vehicleNumber, String ownerName, int insuredAmount)
147     {
148         super(vehicleNumber, ownerName, insuredAmount);
149     }
150
151     @Override
152     public double calculateClaimAmount()
153     {
154         return getInsuredAmount() * 0.80;
155     }
156 }
157
158 class Bike extends Vehicle
159 {
160     public Bike()
161     {
162
163     }
164
165     public Bike(String vehicleNumber, String ownerName, int insuredAmount)
166     {
167         super(vehicleNumber, ownerName, insuredAmount);
168     }
169
170     @Override
171     public double calculateClaimAmount()
172     {
173         return getInsuredAmount() * 0.60;
174     }
175 }
176
177 class Truck extends Vehicle
178 {
179     public Truck()
180     {
181
182     }
183
184     public Truck(String vehicleNumber, String ownerName, int insuredAmount)
185     {
```

```
186         super(vehicleNumber, ownerName, ins  
187     }  
188  
189     @Override  
190     public double calculateClaimAmount()  
191     {  
192         return getInsuredAmount() * 0.90;  
193     }  
194 }
```



Input

3

Car

TN10AB1234



Output

Vehicle : TN10AB1234
Claim Amount : 640000.0

Vehicle : TN09XY5678
Claim Amount : 60000.0



✓ Submitted